

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING
(CYBER SECURITY)

CR23532

CYBER AND DIGITAL FORENSICS

SUBMITTED BY

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EXPERIMENT 1

STUDY OF COMPUTER FORENSICS AND VARIOUS TOOLS USED IN FORENSIC INVESTIGATION

AIM

To gain knowledge about computer forensics, learn the major steps involved in digital forensic investigations, and study the tools commonly used to investigate digital evidence.

SOFTWARE REQUIREMENTS

- **Operating System:** Windows or Linux
- **Digital Forensic Tools:** Autopsy, FTK Imager, The Sleuth Kit (TSK), and other forensic utilities.

THEORY

Computer forensics, commonly referred to as digital forensics, deals with the systematic identification, acquisition, protection, examination, and interpretation of digital evidence obtained from computers and other electronic devices. It plays an important role in examining cybercrimes, security breaches, illegal system access, and other incidents related to digital environments.

Digital forensic experts investigate information available in storage media, operating system records, application logs, web browser activity, deleted files, and various other digital sources. These sources may contain useful information that helps establish the facts related to an incident.

The main purpose of digital forensics is to retrieve and examine digital evidence without compromising its original integrity. It also ensures that the investigation results are properly documented and can be reliably presented for further technical or legal examination.

DIGITAL FORENSICS PROCESS

A digital forensic investigation follows a structured sequence of stages to handle evidence appropriately.

1. Identification

This stage involves locating possible sources of digital evidence, including computers, hard drives, USB storage devices, mobile phones, and system log files.

2. Collection

Relevant evidence is acquired from the identified sources using suitable forensic methods and specialized tools.

3. Preservation

The collected evidence is safeguarded against accidental damage, unauthorized access, or alteration. Investigators may generate forensic copies of storage media and use hash values to check whether the evidence remains unchanged.

4. Examination

The acquired evidence is inspected to discover useful information, such as files, metadata, deleted content, and other digital traces associated with the investigation.

5. Analysis

The information obtained during examination is interpreted to determine connections between events, detect potentially malicious activities, and derive meaningful conclusions.

6. Presentation

The investigation process, evidence examined, observations, and final conclusions are recorded in a structured report for technical assessment or legal proceedings.

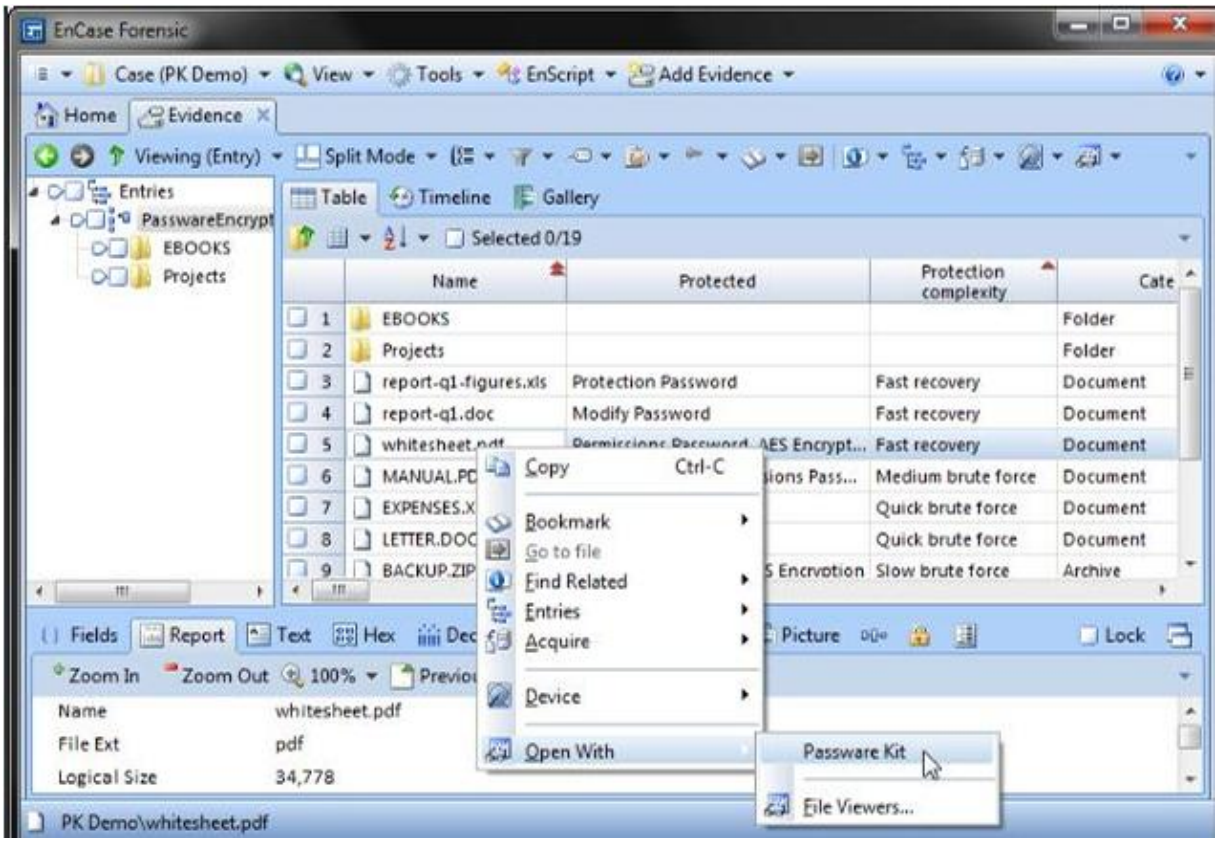
DIFFERENT TOOLS USED IN COMPUTER FORENSICS

Computer forensic investigations make use of both commercial software packages and freely available open-source utilities. Each tool provides specific capabilities for handling and examining digital evidence.

A. EnCase Forensic

EnCase Forensic is a commercially available digital investigation solution designed to acquire, inspect, and analyze electronic evidence. It is commonly used in investigations involving computers and different types of storage media.

EnCase Forensic is a commercial digital forensic investigation tool used to acquire, examine, and analyze digital evidence. It supports forensic investigations involving computers and storage devices.



B. Forensic Toolkit (FTK)

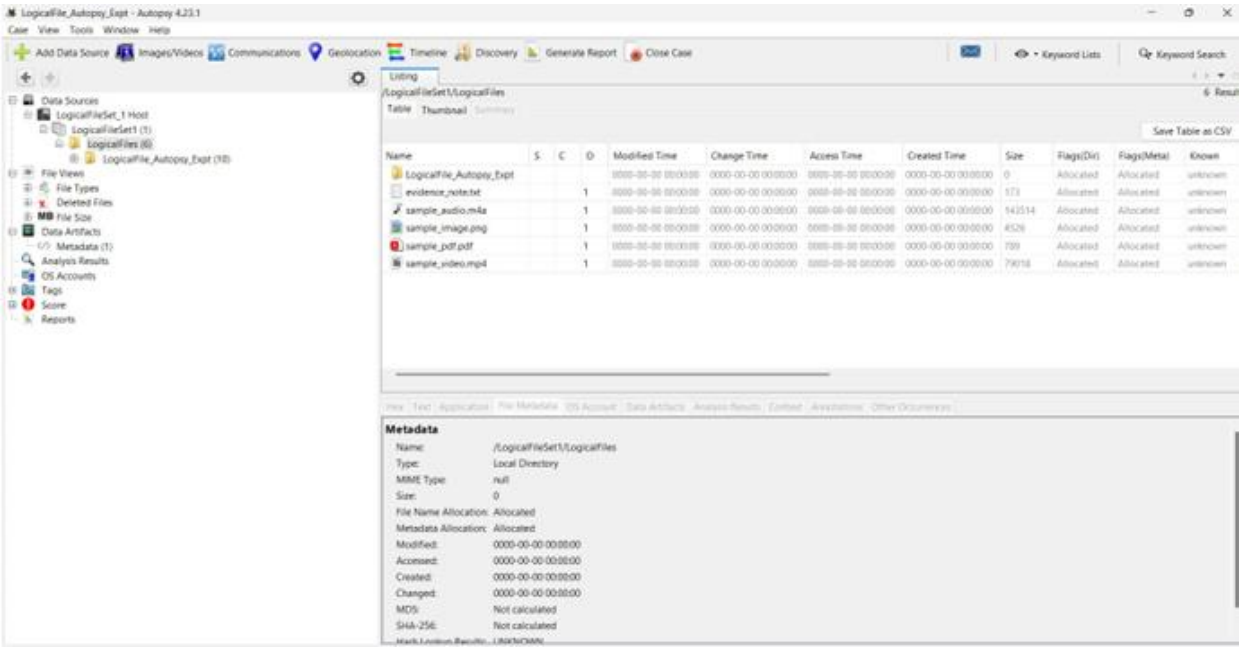
Forensic Toolkit, commonly known as FTK, is a forensic software suite that assists investigators in examining digital evidence. Its features include searching for relevant information, analyzing files, and reviewing data obtained during an investigation.

C. The Sleuth Kit (TSK)

The Sleuth Kit is an open-source set of command-line utilities used to investigate disk images and file systems. It enables investigators to study file-system information, inspect storage structures, and retrieve data that may be important to a forensic examination.

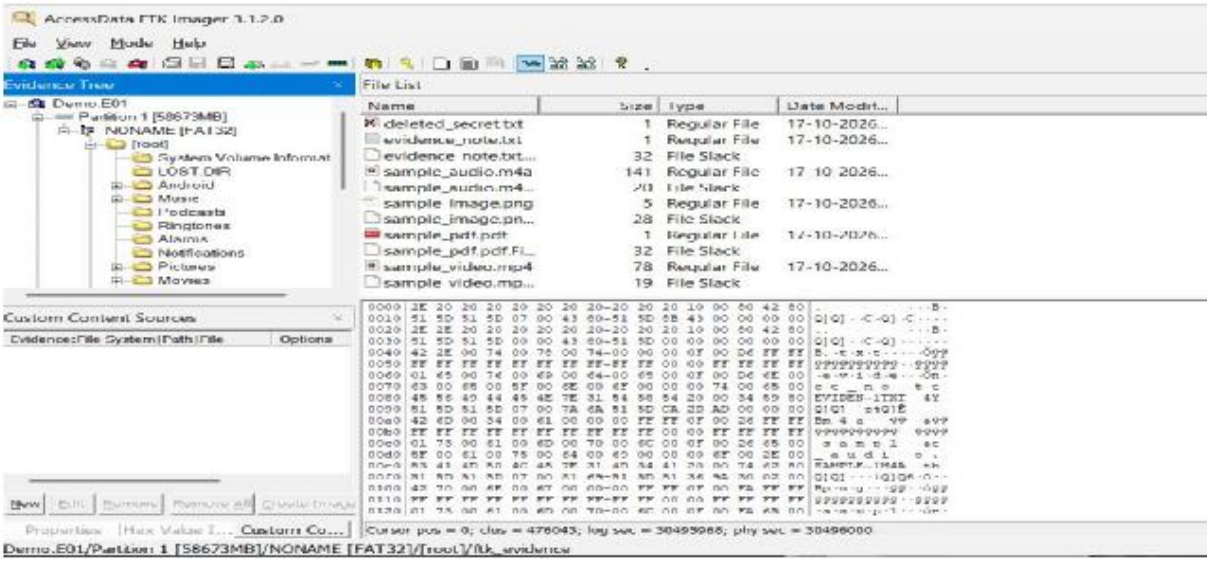
D. Autopsy

Autopsy is a graphical digital forensic application that provides an accessible interface for The Sleuth Kit and other forensic utilities. It offers features for analyzing disk images, recovering files, performing keyword searches, and examining digital artifacts.



E. FTK Imager

FTK Imager is a forensic utility primarily used to create forensic images of storage devices and inspect acquired evidence. It also allows investigators to view supported evidence images and examine the files and folders contained within them.



APPLICATIONS OF COMPUTER FORENSICS

Digital forensics is applied in several fields related to cybersecurity, data analysis, and investigations.

1. Cybercrime Investigation: Digital evidence is examined to investigate unauthorized access, malware-related incidents, and other cyberattacks.

2. Data Recovery: Deleted, damaged, or inaccessible files may be retrieved when recovery is technically possible.

3. Incident Response: Forensic techniques help investigate security incidents, identify relevant evidence, and understand the possible reasons behind an incident.

4. Employee Misconduct Investigation: During authorized workplace investigations, relevant digital records and artifacts can be examined to determine whether inappropriate activities have occurred.

5. Legal Investigations: Digital evidence is collected, preserved, and analyzed to support investigations and potential legal proceedings.

6. Network Investigation: Network logs and associated digital records are studied to identify suspicious communications and understand potentially harmful network activities.

GENERAL PROCEDURE

1. Learn the basic principles and concepts associated with computer and digital forensics.
2. Identify and understand the different phases of a digital forensic investigation.
3. Study the features and uses of commercial and open-source forensic applications.
4. Examine the functions of forensic utilities, including FTK Imager, Autopsy, and The Sleuth Kit.
5. Understand how digital forensics supports cybercrime investigations, incident analysis, and data recovery.
6. Record the observations and summarize the findings obtained from the study.

OBSERVATION

The experiment helped develop an understanding of computer forensics and its significance in identifying and examining digital evidence during cybercrime investigations.

Various forensic applications were studied to understand their individual purposes and capabilities. The different phases of the forensic investigation process, including identification, collection, preservation, examination, analysis, and presentation, were also understood.

RESULT

Thus, computer forensics, its investigation process, and the various tools used for digital forensic examination were studied successfully.

EXPERIMENT 2

LIVE FORENSICS CASE INVESTIGATION USING AUTOPSY

AIM

To carry out a digital forensic investigation with Autopsy by setting up a new case, importing a data source, processing digital evidence, and inspecting the forensic artifacts obtained during the investigation.

SOFTWARE REQUIREMENTS

- **Operating System:** Windows
- **Software:** Autopsy Digital Forensic Tool
- **Evidence Source:** A suitable forensic image or disk image
- **Hardware:** A computer with adequate memory and storage capacity

THEORY

Autopsy is a digital forensic platform that offers a graphical user interface for investigating and analyzing digital evidence. It incorporates The Sleuth Kit and other forensic utilities to examine files, disk images, and digital artifacts.

The tool provides several capabilities, including file-system examination, keyword-based searches, retrieval of deleted files, and analysis of digital traces associated with user activities.

During a forensic investigation, the examiner first creates a case to maintain the investigation details and organize the related evidence. A suitable data source, such as a disk image, is subsequently imported into the case. Autopsy examines the selected source through the configured ingest modules and presents the processed information for further analysis.

The extracted results may contain useful details such as files, user account information, web browser records, metadata, and other artifacts. These findings assist investigators in identifying information relevant to the forensic examination.

PROCEDURE

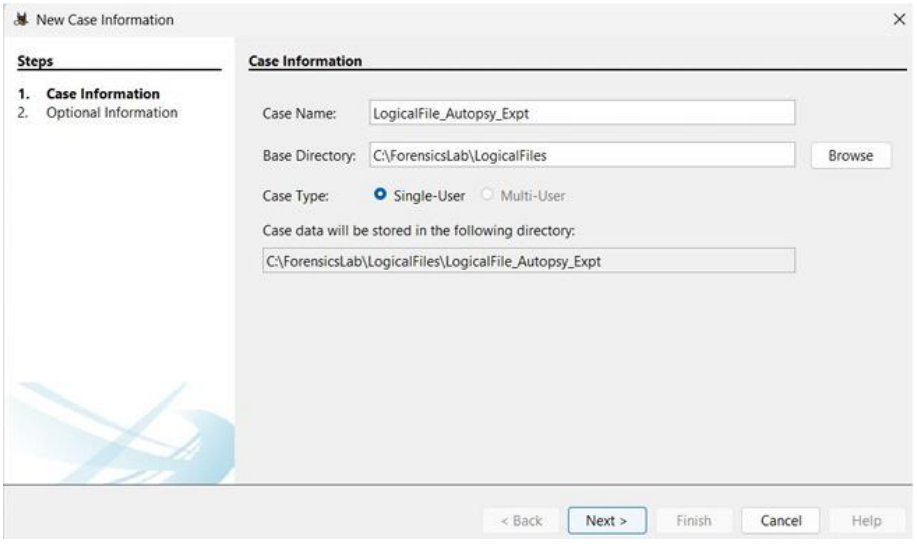
Step 1: Launch Autopsy

1. Download and install Autopsy using its official website.
2. Start the Autopsy digital forensic application.
3. Choose the **New Case** option from the welcome screen to begin the investigation.



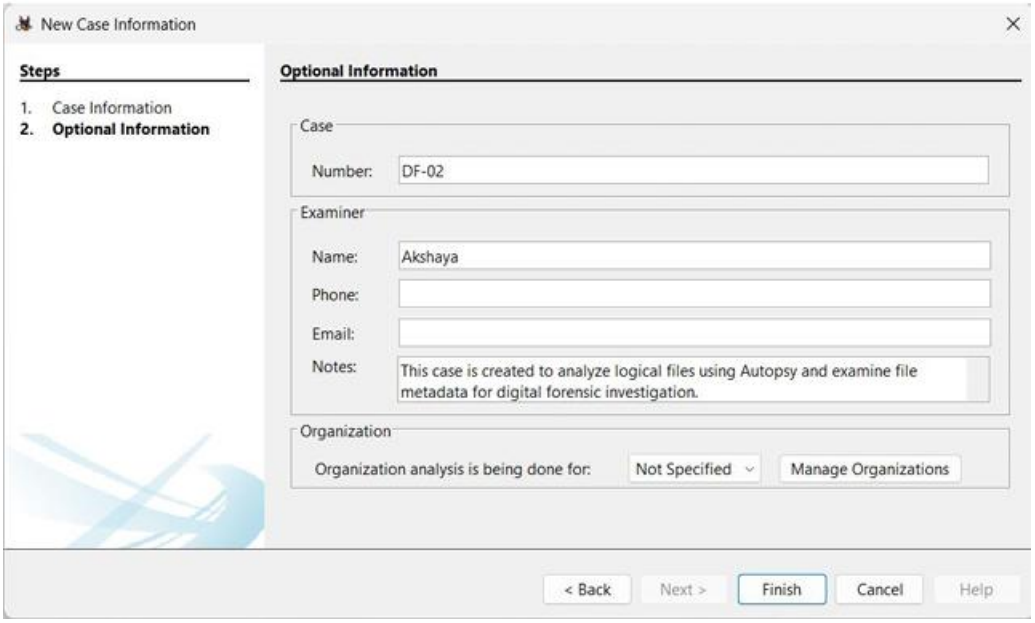
Step 2: Create a New Case

1. Provide an appropriate name for the forensic case.
2. Select the base directory in which the case-related files and information will be saved.
3. Click **Next** to move to the following stage.



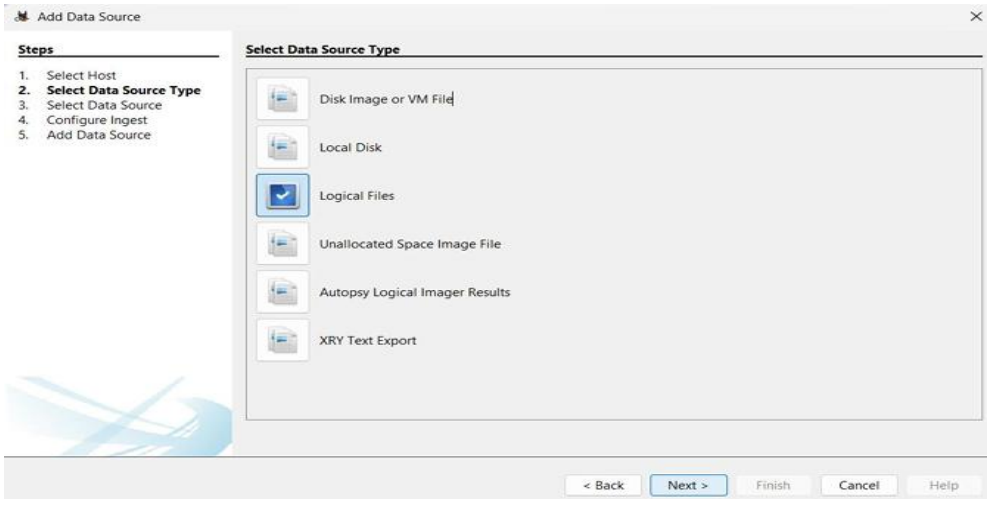
Step 3: Provide Case Details

1. Enter the identification number assigned to the case.
2. Fill in the examiner's information.
3. Supply any other relevant details requested by the application.
4. Click **Finish** to complete the case creation process.



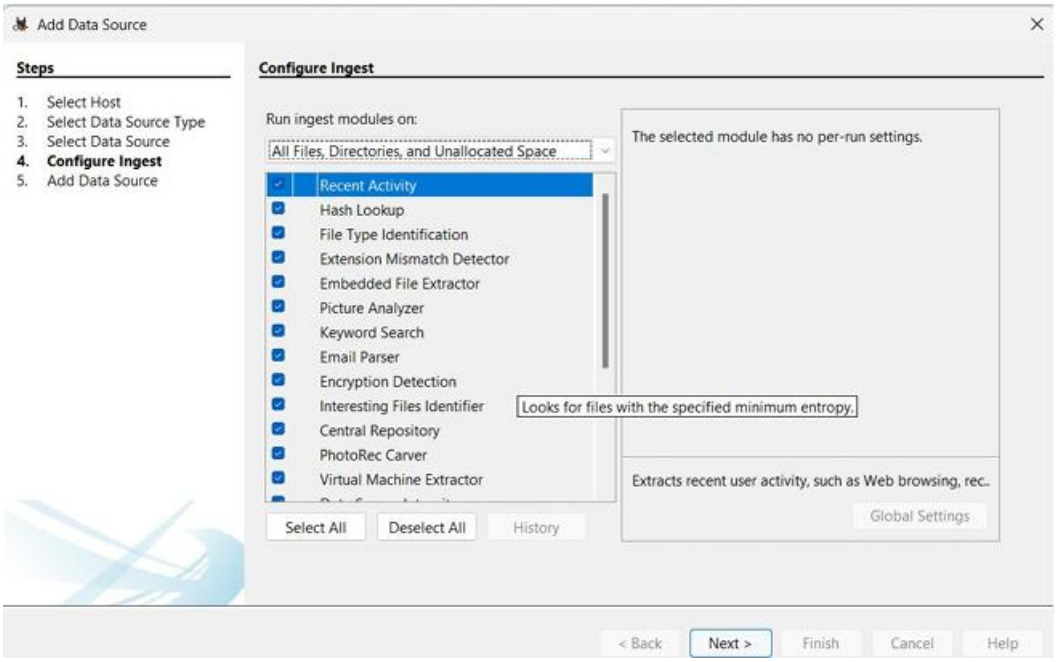
Step 4: Add a Data Source

1. Once the case has been created, choose the option for adding a data source.
2. Select the suitable source category, such as a disk image or a supported virtual machine file.
3. Navigate to the folder containing the forensic image.
4. Select the required image file and continue to the next stage.



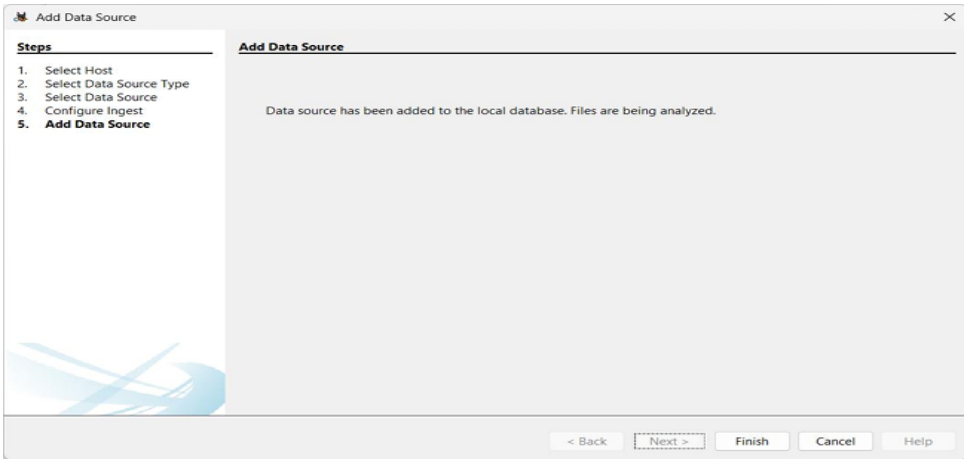
Step 5: Select and Configure Ingest Modules

1. Examine the ingest modules available in Autopsy.
2. Enable the modules that are appropriate for the evidence and investigation requirements.
3. Click **Next** to proceed.
4. Check the selected configurations and finish adding the data source.



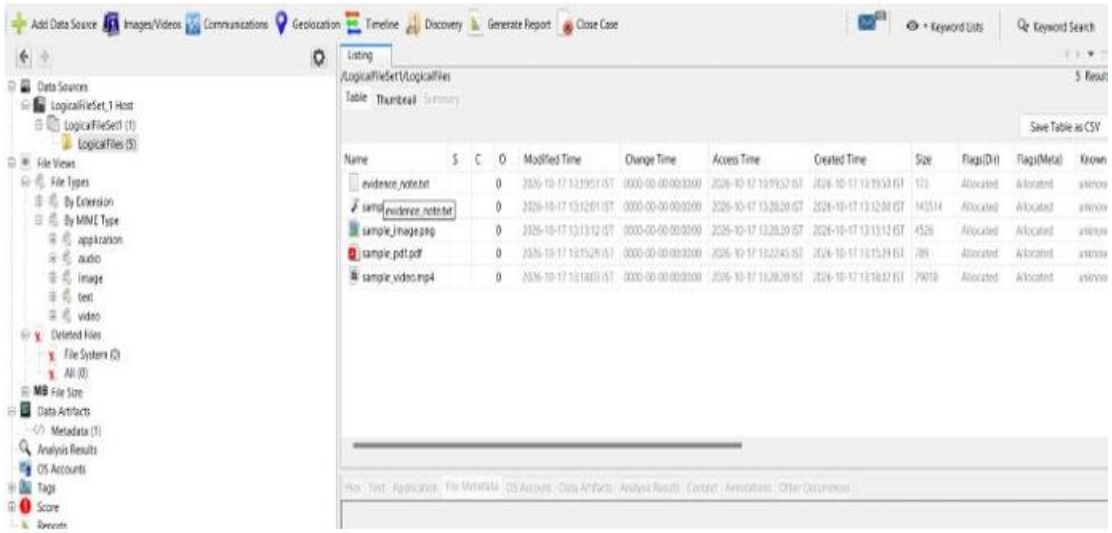
Step 6: Process the Evidence Source

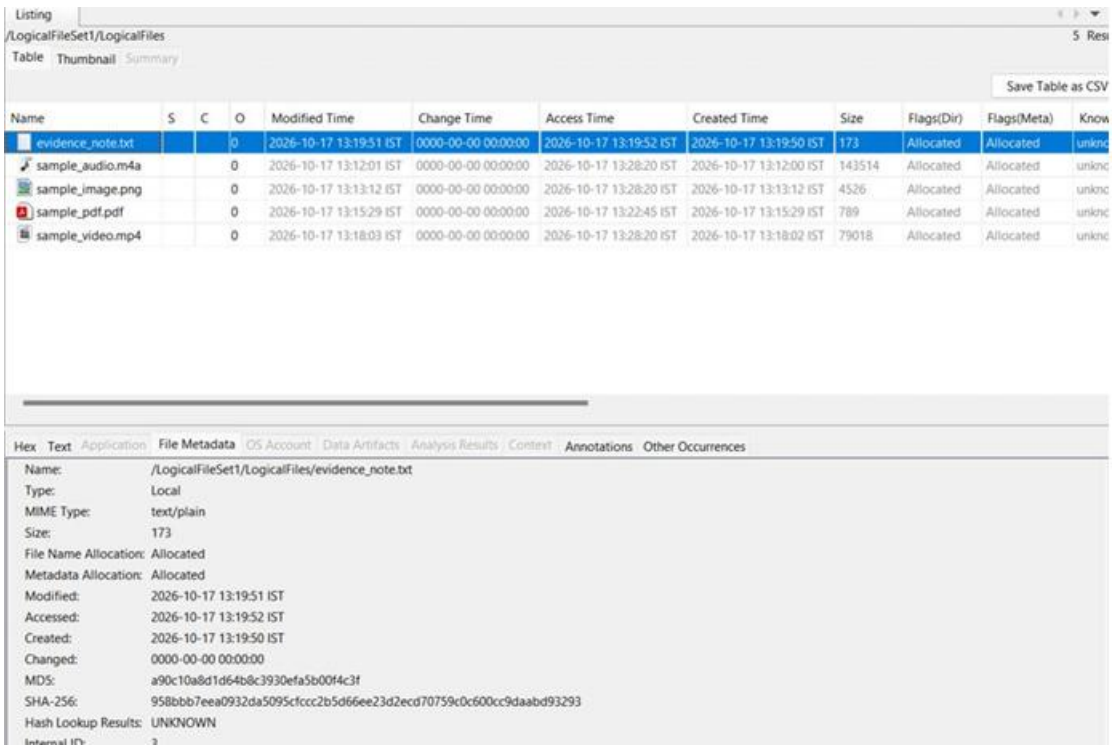
1. Complete the data-source addition wizard.
2. Allow Autopsy to analyze and process the selected forensic image.
3. Wait until the processing tasks have completed.
4. Review the information and results displayed in the case interface.



Step 7: Examine the Forensic Artifacts

After processing is complete, inspect the artifacts generated by Autopsy. Depending on the contents of the evidence image and the ingest modules enabled, the available results may include files, deleted data, browser-related information, metadata, and other digital records.





VERIFICATION OF EVIDENCE INTEGRITY USING SHA-256

After the logical files were processed in Autopsy, the integrity of the evidence was cross-checked using the SHA-256 hashing algorithm through the Windows Command Prompt. The generated hash values can be compared with the corresponding reference values to determine whether the examined data remains unchanged.



OBSERVATION

A forensic case was successfully created in Autopsy, and a suitable data source was added for examination. The configured ingest modules processed the selected evidence and produced digital artifacts for further inspection.

The case interface displayed the files and other information identified during processing. The type and quantity of artifacts available for examination depended on the data contained in the selected forensic image and the modules used.

RESULT

Thus, the digital forensic investigation procedure using Autopsy was studied and carried out by creating a case, importing a data source, processing the evidence, examining the resulting artifacts, and understanding the verification of evidence integrity using SHA-256.

EXPERIMENT 3

RECOVERY OF DELETED FILES USING FORENSIC TOOLS

AIM

To retrieve deleted files from a forensic image with the help of digital forensic utilities and confirm the integrity of the recovered information.

SOFTWARE REQUIREMENTS

- **Operating System:** Windows
- **Forensic Tool:** FTK Imager
- **File Recovery Software:** Recover MyFiles
- **Evidence Source:** USB storage device or another suitable test data source

THEORY

Deleting a file from a storage device does not necessarily remove all its data permanently. In many file systems, the deletion process marks the storage space occupied by the file as available for reuse rather than immediately erasing its contents. Therefore, the deleted data may remain recoverable until new information overwrites the same storage area.

Digital forensic applications are used to inspect storage media and forensic images to identify and retrieve deleted files. FTK Imager assists in creating and examining forensic images, whereas specialized recovery utilities can search for deleted files and other data that may still be available.

A forensic image provides a copy of the source data as it existed during acquisition. This enables investigators to examine the captured information without repeatedly working directly on the original storage device, helping preserve the source evidence during analysis.

PROCEDURE

Step 1: Prepare a Sample File

1. Launch Notepad on the Windows system.
2. Enter some sample text that will be used to test the recovery process.
3. Save the document as deleted_secret.txt on a test USB drive or another suitable test location.

```
CONFIDENTIAL – FORENSIC EXPERIMENT FILE

Case Reference: CF-2026-001
Project: Cyber and Digital Forensics
Prepared By: Student Name
Date: 17-10-2026

Subject: Sample confidential project document

This file contains simulated evidence for the Autopsy and FTK Imager
laboratory experiments.

Employee ID: EMP-10293
Username: student_forensics
Email: student.forensics@example.com
Project Code: CYBER-DFIR-2026
Access Level: Restricted

Confidential project notes:
The internal audit identified unauthorized access attempts on the
company research server. The suspect may have transferred project
documents to a USB storage device.

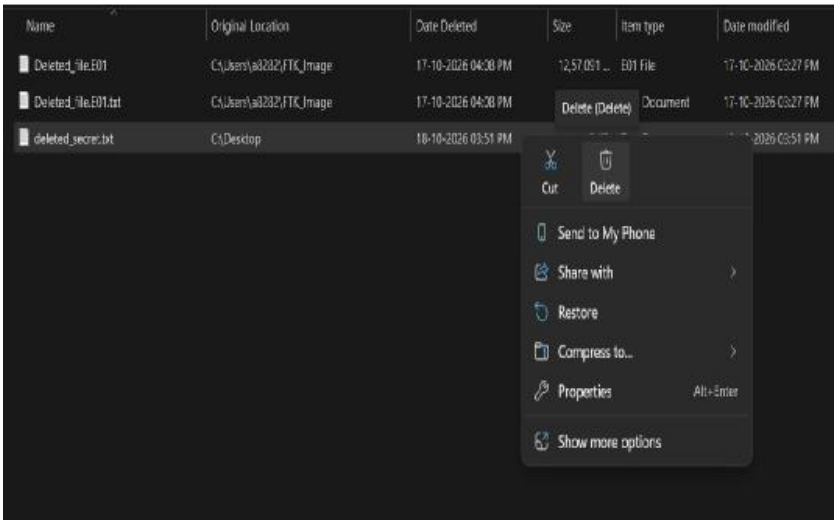
Important keywords:
confidential
unauthorized access
USB transfer
project documents

Password Hint: ForensicsLab@2026

Disclaimer:
This file contains fictional information created only for educational
and forensic analysis purposes.
```

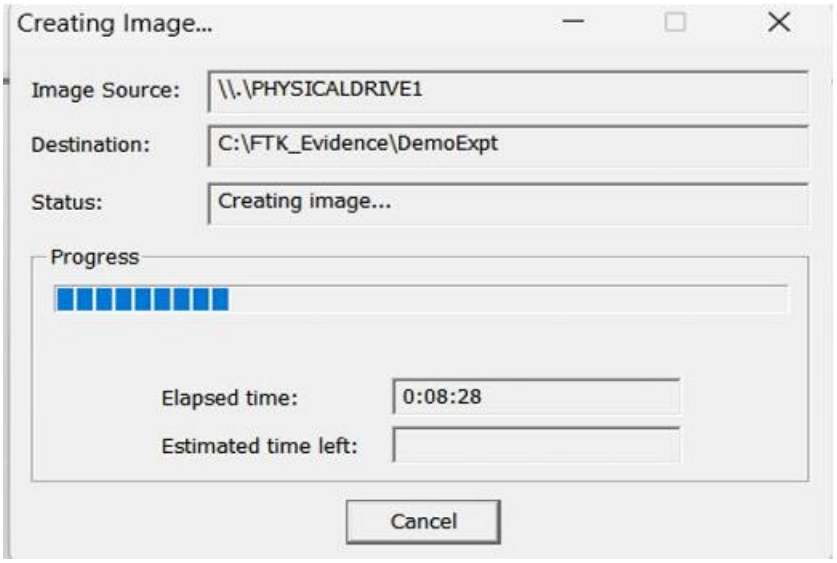
Step 2: Delete the Created File

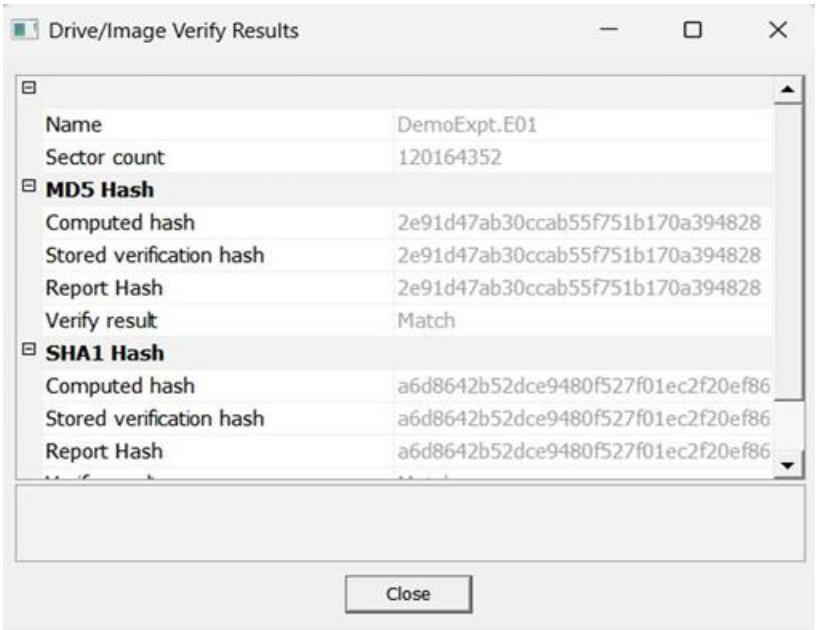
1. Navigate to the location where deleted_secret.txt was saved.
2. Right-click the file and choose the **Delete** option.
3. Verify that the file is no longer displayed in its original location.

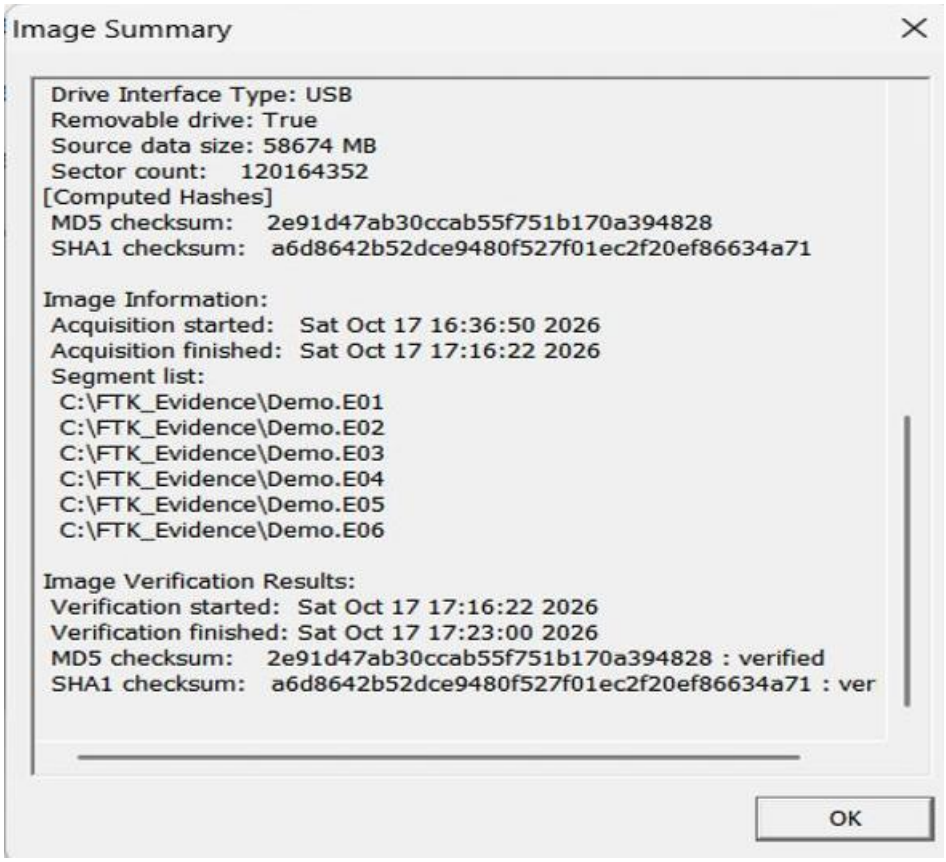


Step 3: Acquire a Forensic Image Using FTK Imager

1. Start the FTK Imager application.
2. Navigate to **File → Create Disk Image**.
3. Select the appropriate type of evidence source.
4. Choose the test storage device that previously contained the deleted file.
5. Enter the necessary case details and evidence information.
6. Choose the folder where the image will be saved and select the required image format.
7. Begin the imaging operation and allow the process to finish.

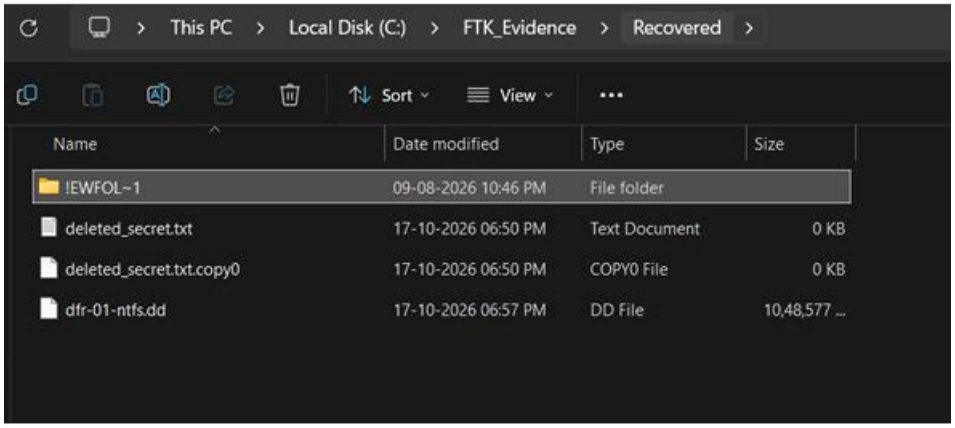






Step 4: Validate the Recovered File

1. Open the file retrieved through the recovery process.
2. Examine the file to determine whether the previously entered sample text is available.
3. Compare the recovered data with the original content to check whether the information has been restored correctly.



OBSERVATION

A sample text document was prepared and deleted from a test storage location. FTK Imager was then used to create a forensic image of the selected storage source, which was examined with a file recovery utility.

The recovery results were analyzed to establish whether the deleted document and its contents could be retrieved. The success of the recovery process depended on factors such as the file system, the method used to acquire the image, and whether the storage space containing the deleted data had been overwritten.

RESULT

Thus, the process of recovering deleted files using digital forensic tools was studied and carried out. The forensic image was examined for recoverable deleted data, and the recovered file contents were checked against the original sample text wherever recovery was successful.

EXPERIMENT 4

IDENTIFICATION OF THE LAST CONNECTED USB DEVICE USING USB FORENSICS

AIM

To determine which USB devices have been connected to a computer previously and analyze the available device information using the USBDeview forensic utility.

SOFTWARE REQUIREMENTS

- **Operating System:** Windows
- **Tool:** USBDeview
- **System Requirement:** A computer containing records of previous USB device connections

THEORY

USB forensics is a branch of digital forensics that focuses on examining information related to USB devices connected to a computer. Such records can assist investigators in identifying devices that have been used with a system and understanding their associated connection details.

USBDeview is a utility designed for Windows that provides information about USB devices currently connected to the computer as well as devices that were connected in the past. The tool displays several details that may be useful during a forensic investigation, including:

- **Device Name and Description:** Identifies the USB device and provides a description of its function or type.
- **Device Type:** Indicates the category of the connected USB device.
- **Serial Number:** Displays the device's serial number when the information is available.
- **Connection Date and Time:** Shows the available date and time records associated with the device's connection or registration in the system.
- **Vendor ID (VID):** Represents the identifier associated with the device manufacturer.
- **Product ID (PID):** Identifies the particular product or device model associated with the vendor.

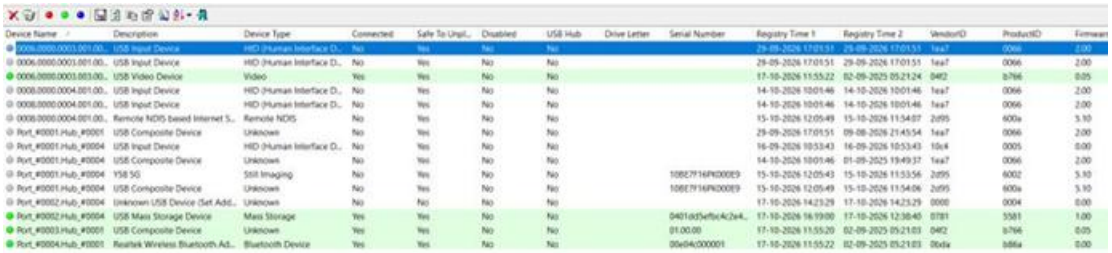
By examining these details, investigators can review USB connection history and identify records relevant to a digital forensic investigation.

PROCEDURE

1. Download and open the USBDeview utility on the Windows computer.
2. Allow the application to retrieve and display the available USB device records.
3. Review the entries corresponding to currently connected devices and those previously connected to the system.
4. Locate the required USB device by examining its name and description.
5. Inspect the device's serial number, Vendor ID (VID), and Product ID (PID), wherever these details are available.
6. Review the date and time information displayed for the selected device.
7. Document the relevant device information for further forensic analysis.
8. Take screenshots of the USB device listing and the detailed information of the selected device for documentation.

OBSERVATION

USBDeview presented records of USB devices associated with the Windows system, including information about devices that had been connected previously. The available details, such as device descriptions, identification numbers, serial numbers, and connection-related timestamps, could be examined to identify relevant USB device records.





RESULT

Thus, USBDeview was used to examine the USB connection history of the computer and inspect the available device details for digital forensic investigation.

EXPERIMENT 5

EXAMINATION OF RECENT COMPUTER ACTIVITY USING LASTACTIVITYVIEW

AIM

To investigate recent user activities and system events recorded on a Windows computer with the help of the LastActivityView forensic utility.

SOFTWARE REQUIREMENTS

- **Operating System:** Windows
- **Tool:** LastActivityView

THEORY

LastActivityView is a Windows-based utility that gathers information from various sources within the operating system and presents the available records of user activities and system events in a consolidated log.

The utility assists forensic investigators in reviewing activity-related information and examining certain actions performed on a computer. The displayed records may include system events and details associated with recent user activity. The amount and type of information available depend on the records maintained by the operating system and the data present on the computer.

By analyzing these entries, investigators can obtain useful insights into selected aspects of the computer's recent activity and identify records that may be relevant to a digital forensic examination.

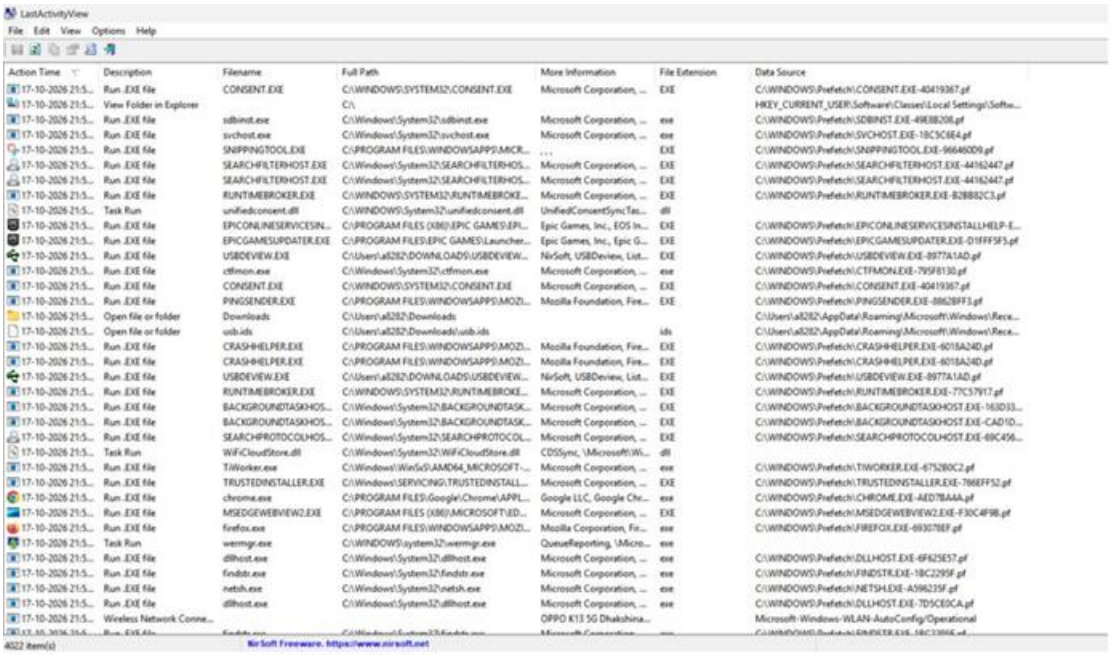
PROCEDURE

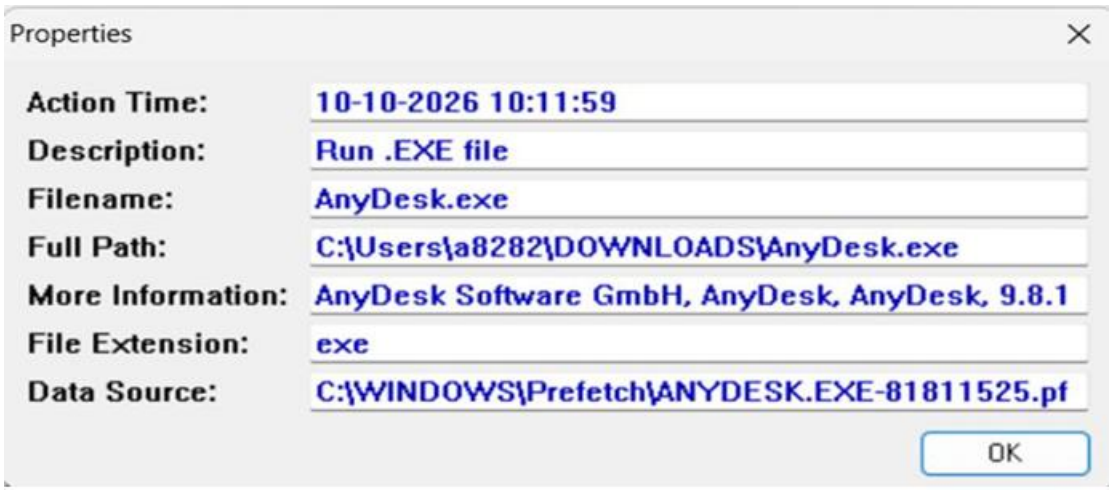
1. Download and start the LastActivityView utility on the Windows computer.
2. Allow the application to collect information from the available system records.
3. Examine the activity entries presented in the application's main window.
4. Review the available information, including the activity description, associated filename or application, and recorded date and time, wherever provided.
5. Locate entries that are relevant to the computer's recent activities.
6. Document the observations for further forensic examination.

7. Take screenshots of the activity log and important entries as supporting evidence.

OBSERVATION

LastActivityView displayed the available records of user activities and system events associated with the Windows computer. The entries provided information that could be reviewed to understand certain aspects of the system's recent usage and activity history.





RESULT

Thus, LastActivityView was used to inspect the available records of recent computer activities and system events for digital forensic investigation.

EXPERIMENT 6

EXTRACTION AND ANALYSIS OF BROWSER ARTIFACTS

AIM

To retrieve and analyze web browser artifacts, such as cached data and browsing history, with the help of ChromeCacheView and ChromeHistoryView utilities.

SOFTWARE REQUIREMENTS

- **Operating System:** Windows
- **Web Browser:** Google Chrome
- **Tools:** ChromeCacheView and ChromeHistoryView

THEORY

Browser artifacts refer to the information created or stored by a web browser during internet browsing activities. These records can provide valuable evidence and assist investigators in understanding browser-related activities during digital forensic examinations.

ChromeCacheView is a utility designed to read the cache directory of Google Chrome and display information about the cached content. Depending on the available data, it can present details such as the associated URL, file type, cache filename, file size, and access time.

ChromeHistoryView is a utility that retrieves and displays browsing history records maintained by Google Chrome. The information presented may include the website URL, page title, number of visits, and the time of the most recent visit, depending on the available records.

Both utilities support digital forensic investigations by allowing examiners to inspect browser cache data and browsing history for information relevant to an investigation.

PROCEDURE

A. EXAMINATION USING CHROMECACHEVIEW

1. Download and open the ChromeCacheView utility on the Windows computer.
2. Allow the application to identify and access the available Google Chrome cache records.
3. Inspect the cached file entries displayed in the main application window.
4. Review the available information, including URLs, file types, cache filenames, file sizes, and access times.
5. Select relevant cache entries for further examination.
6. Take a screenshot of the displayed cached file information for documentation.

B. EXAMINATION USING CHROMEHISTORYVIEW

1. Download and start the ChromeHistoryView utility on the Windows computer.
2. Allow the application to locate and retrieve the available Google Chrome browsing history.
3. Review the history entries presented in the application's main window.
4. Examine the available details, such as website URLs, page titles, visit counts, and the most recent visit times.
5. Identify browsing history entries that may be relevant to the forensic investigation.
6. Capture a screenshot of the extracted browsing history records for documentation.

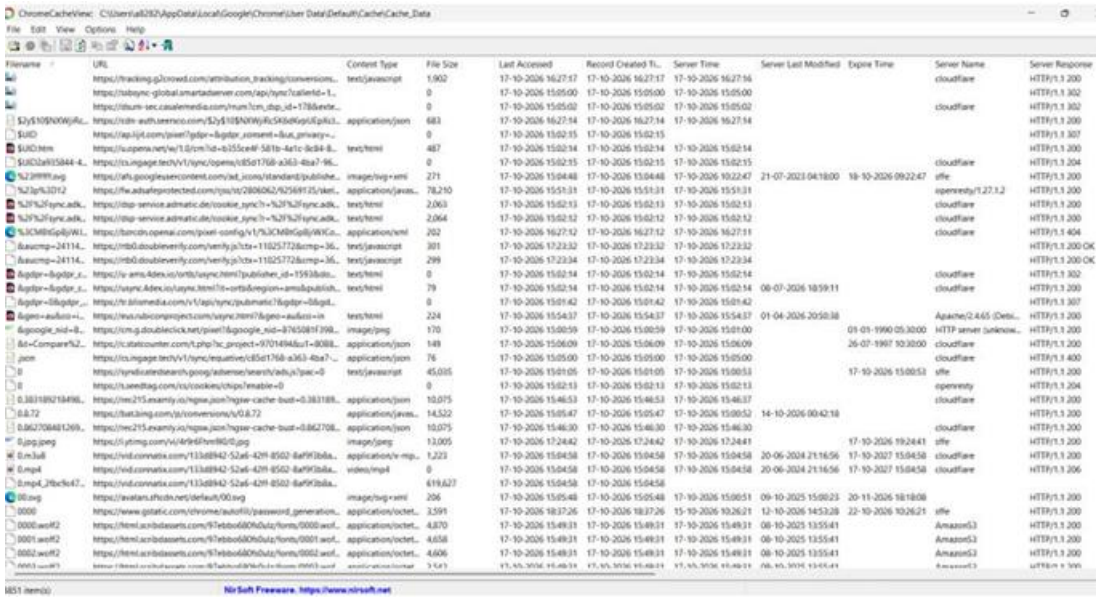
OBSERVATION

The two utilities were used to inspect separate categories of browser artifacts. ChromeCacheView provided information about cached files, while ChromeHistoryView displayed available website browsing records.

S. No.	Tool Name	Artifact Examined	Observation
1	ChromeCacheView	Cached files	Details of the available cached files were displayed for examination.
2	ChromeHistoryView	Browsing history	Available records of previously visited websites were presented for analysis.

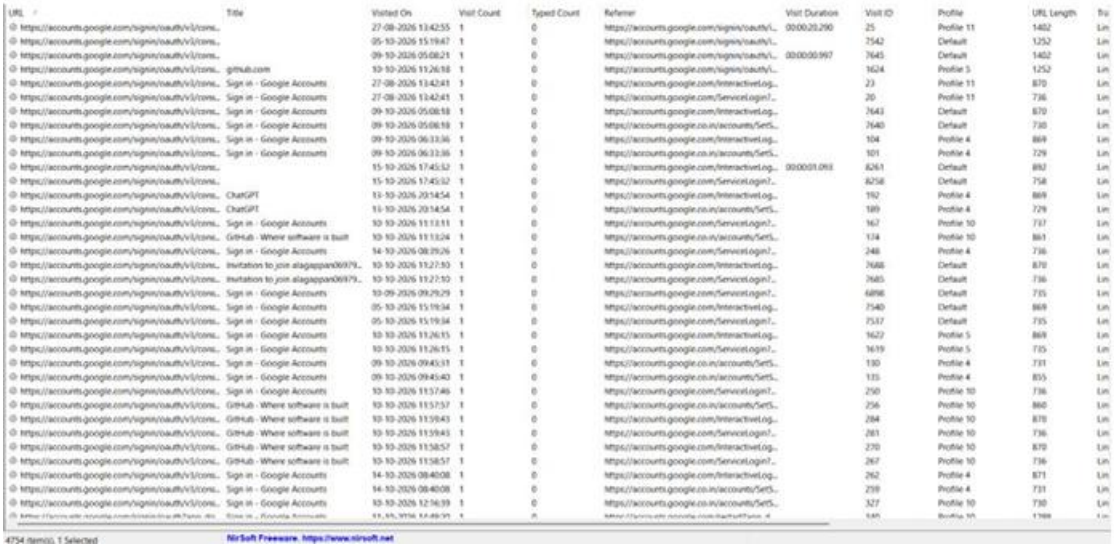
DETAILS OBSERVED USING CHROMECACHEVIEW

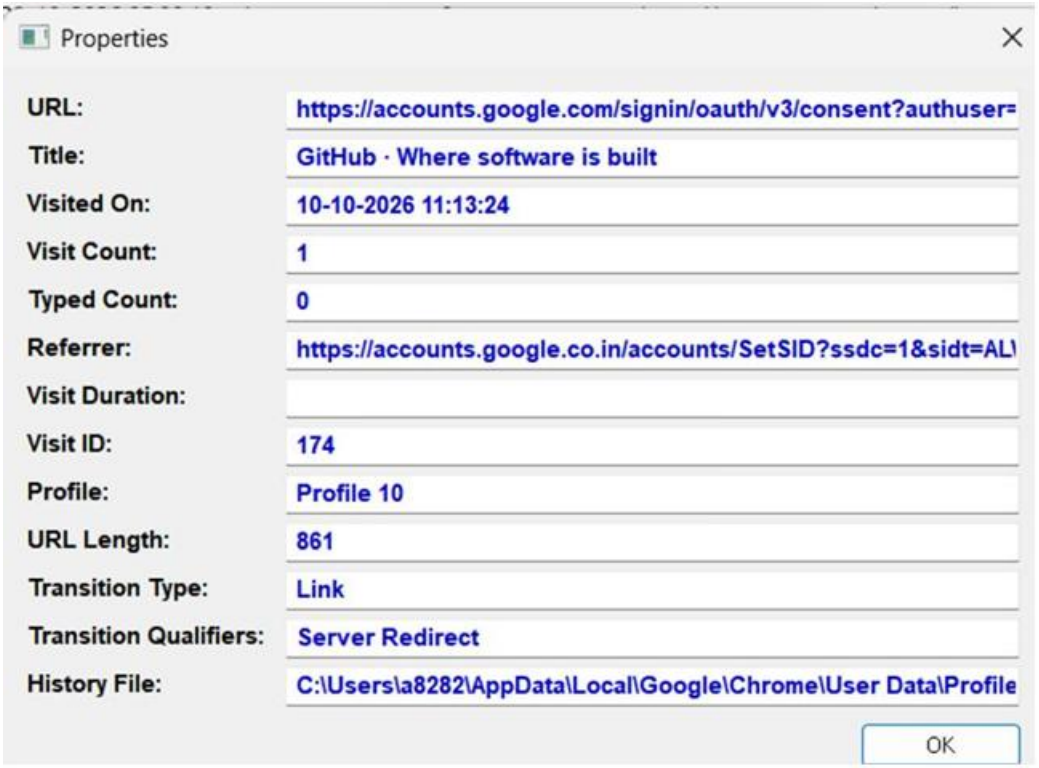
The cached file records were examined to identify the available URLs, file types, filenames, file sizes, and access-time information.



DETAILS OBSERVED USING CHROMEHISTORYVIEW

The browsing history entries were reviewed to examine website URLs, page titles, visit counts, and last-visit timestamps wherever the information was available.





RESULT

Thus, browser artifacts were successfully examined using ChromeCacheView and ChromeHistoryView. The available cached file information and browsing history records were retrieved and analyzed as part of the digital forensic investigation.

EXPERIMENT 7

COMPARISON OF TWO FILES FOR FORENSIC INVESTIGATION USING COMPARE IT SOFTWARE

AIM

To examine and compare two files with the help of Compare It software in order to detect differences in their contents for digital forensic analysis.

SOFTWARE REQUIREMENTS

- **Operating System:** Windows
- **Software:** Compare It
- **Input:** Two sample files for comparison

THEORY

File comparison is an important technique in digital forensics that involves analyzing two files to determine the differences between their contents. This process helps investigators establish whether a document has undergone modifications or whether the information contained in two files varies.

Compare It is a file comparison utility that presents two files alongside each other and uses colour highlighting to indicate differences. It enables users to locate content that has been inserted, removed, or altered between the files.

The results obtained through file comparison can assist forensic investigators in examining document changes, identifying possible modifications, and understanding variations between different versions of a file.

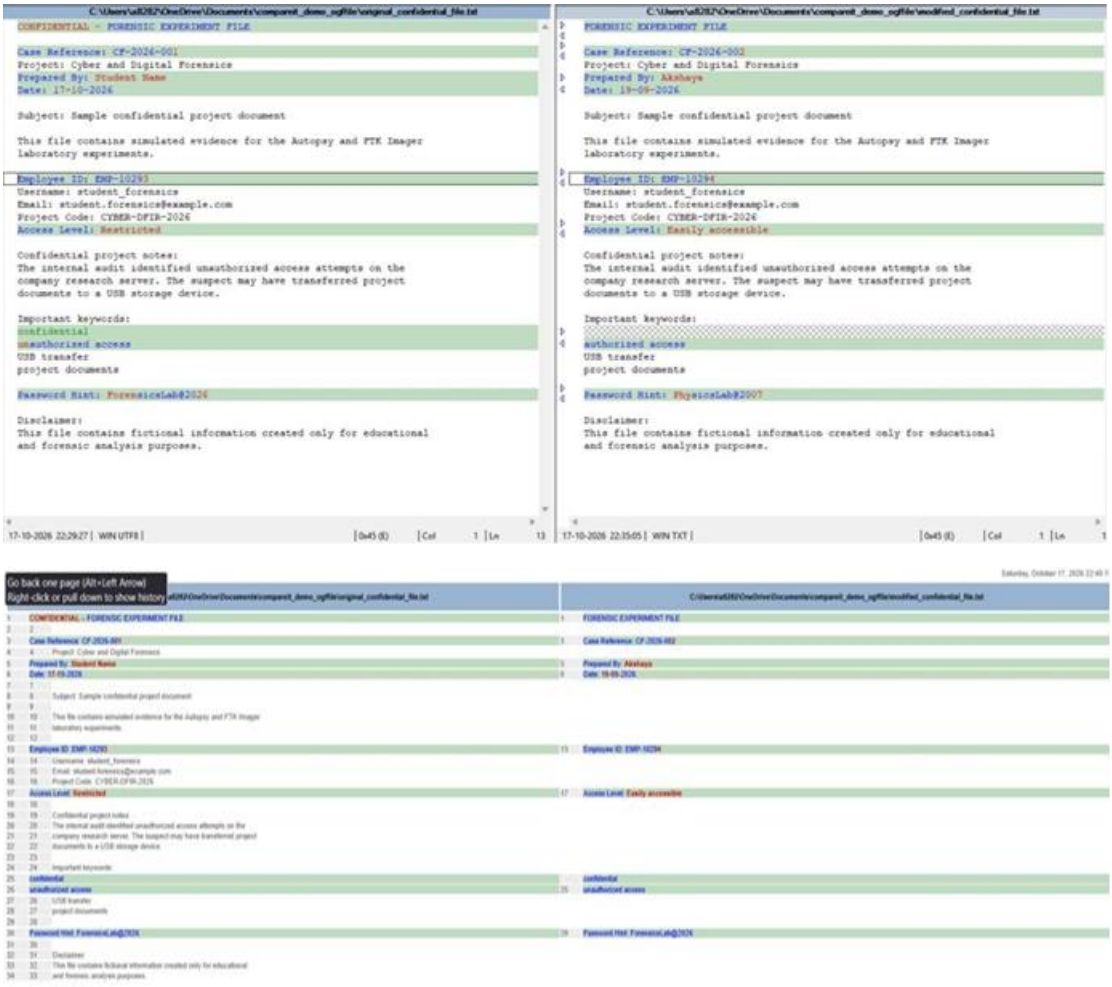
PROCEDURE

1. Download and install the Compare It application on the Windows computer.
2. Create or select two sample files containing similar content with a few intentional differences.
3. Open the Compare It software.
4. Choose the first sample file and click **Open**.
5. Select the second sample file and open it in the application.
6. Allow the software to present the two files in a side-by-side view.

7. Inspect the highlighted portions to locate differences in the file contents.
8. Identify any text that has been inserted, removed, or modified between the two files.
9. Document the findings and take a screenshot of the comparison window for reference.

OBSERVATION

Compare It presented the selected files next to each other and visually marked the variations found in their contents. The highlighted sections helped distinguish differences between the two files, including added, deleted, or modified text.



RESULT

Thus, two sample files were examined using Compare It software, and the differences between their contents were identified for digital forensic investigation.

EXPERIMENT 8

COLLECTION OF EMAIL EVIDENCE FROM THE VICTIM'S PC

AIM

To obtain email-related digital evidence from a computer by acquiring its RAM using DumpIt and analyzing the captured memory for information relevant to the investigation.

SOFTWARE REQUIREMENTS

- **Operating System:** Windows
- **Tool:** DumpIt
- **Additional Requirement:** Memory analysis tools or scripts mentioned in the laboratory procedure

THEORY

Email forensics is the process of locating and analyzing email-associated information available on a computer system. Such information can provide useful insights into communication-related activities and support the investigation of digital incidents.

DumpIt is a memory acquisition utility used to capture the physical memory of a Windows computer and save it as a memory dump. Since RAM is volatile, it may contain temporary information related to currently running applications, including traces of email activity performed through a web browser.

Once the memory image has been acquired, suitable forensic analysis tools can be used to examine its contents and search for potentially relevant artifacts. These may include email addresses and other information associated with email-related activities.

Maintaining the integrity of the acquired memory image is essential throughout the investigation. The dump and the results obtained from its analysis should be handled carefully and preserved for documentation and further examination.

PROCEDURE

A. ACQUISITION OF RAM USING DUMPIT

1. Download and prepare the DumpIt utility on the Windows computer.
2. Execute DumpIt.exe with the necessary administrative privileges.
3. Carefully review the confirmation message displayed by the application.

4. Type **Y** to approve and initiate the memory acquisition, as specified in the lab manual.
5. Allow the utility to complete the RAM acquisition process.
6. Navigate to the output directory and locate the generated raw memory dump.
7. Note down the filename and size of the acquired memory dump for documentation.

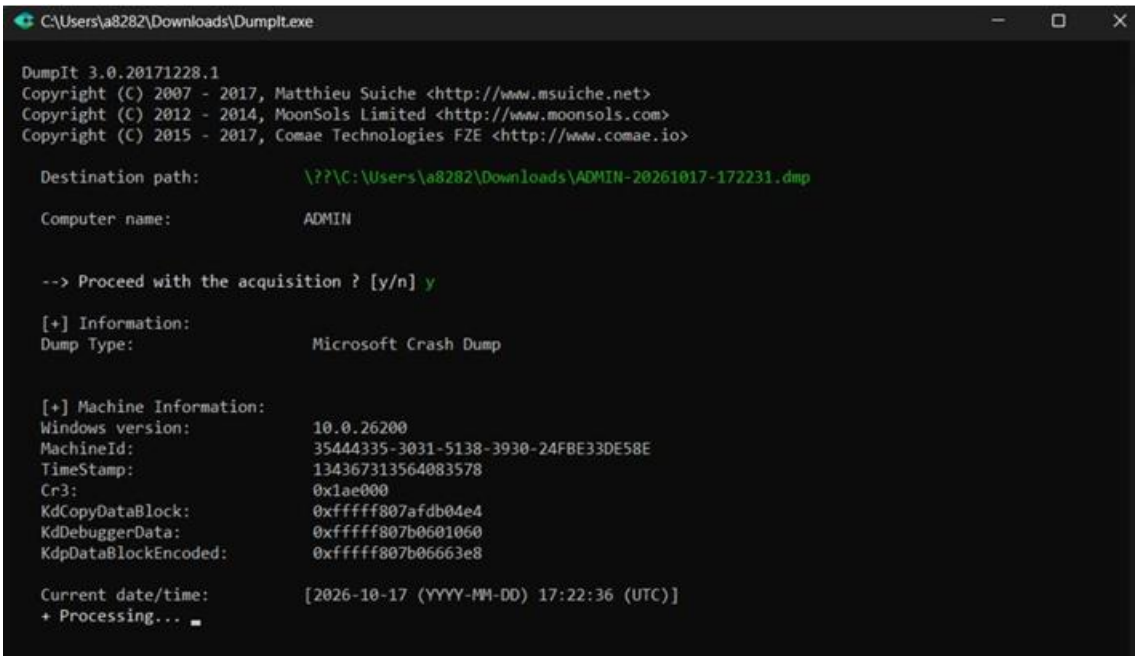
B. EXAMINATION OF EMAIL-RELATED EVIDENCE

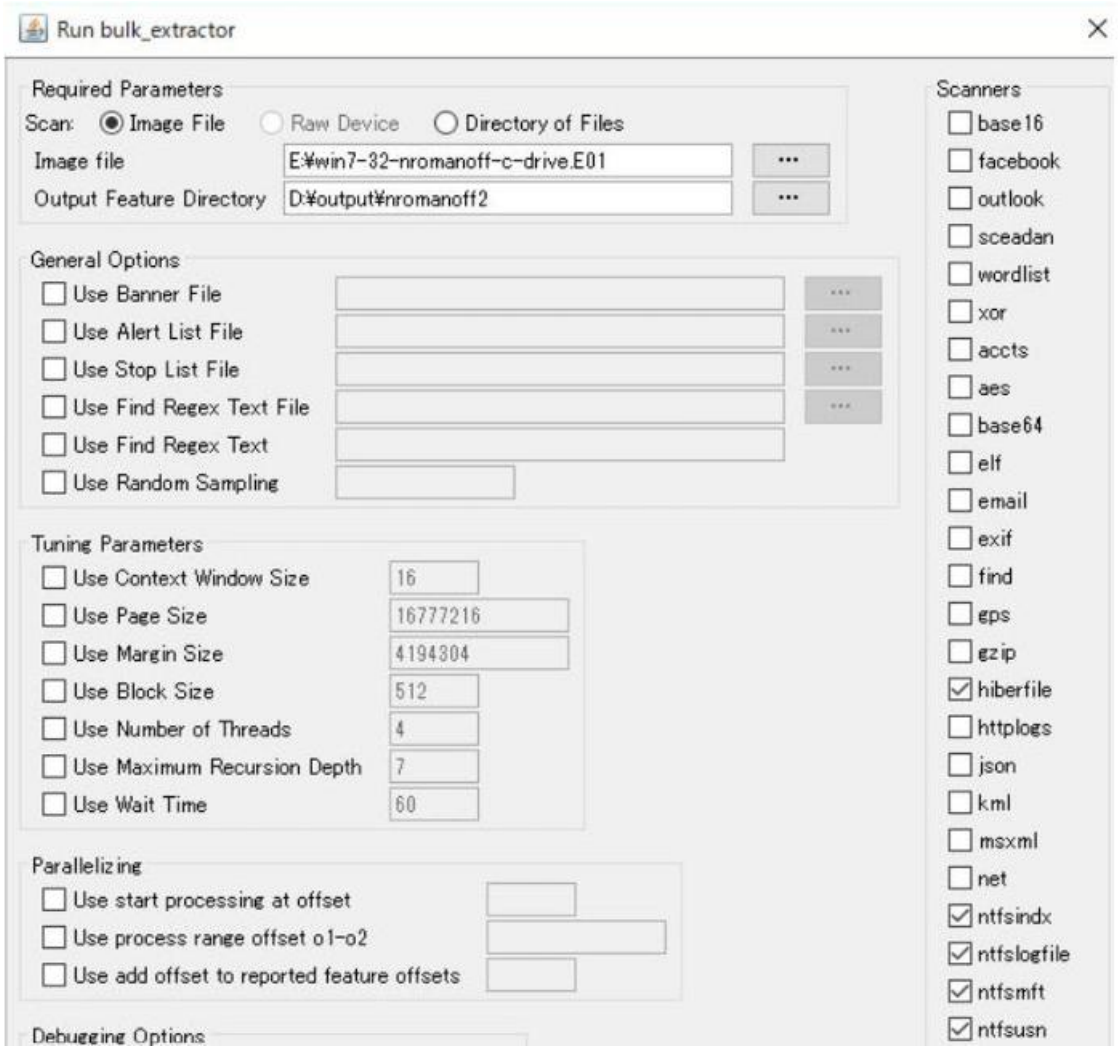
1. Load the acquired memory dump into the memory examination environment specified in the laboratory procedure.
2. Analyze the available output to locate information associated with email activity.
3. Open email.txt and review the email-related details extracted by the prescribed procedure.
4. Examine email_histogram.txt to review the summary or histogram of the identified email-related information.
5. Document the relevant findings obtained from the generated output files.
6. Store the memory dump and the corresponding analysis results securely for future reference and forensic documentation.

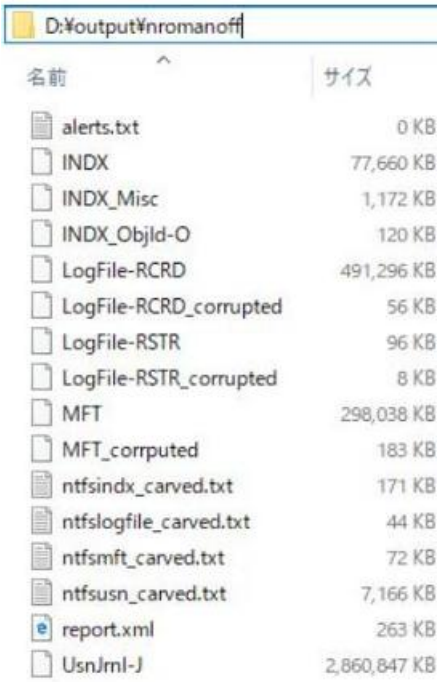
OBSERVATION

The procedure involved acquiring the computer's RAM using DumpIt and examining the resulting memory dump for email-related information. The generated output files, email.txt and email_histogram.txt, were designated for reviewing the extracted details and summary information.

The availability of email-related artifacts depended on the contents of the captured memory and the analysis method used.


C:\Users\a8282\Downloads\DumpIt.exe
DumpIt 3.0.20171228.1
Copyright (C) 2007 - 2017, Matthieu Suiche <http://www.msuiche.net>
Copyright (C) 2012 - 2014, MoonSols Limited <http://www.moonsols.com>
Copyright (C) 2015 - 2017, Comae Technologies FZE <http://www.comae.io>
Destination path: \??\C:\Users\a8282\Downloads\ADMIN-20261017-172231.dmp
Computer name: ADMIN
--> Proceed with the acquisition ? [y/n] y
[+] Information:
Dump Type: Microsoft Crash Dump
[+] Machine Information:
Windows version: 10.0.26200
MachineId: 35444335-3031-5138-3930-24FBE33DE58E
TimeStamp: 134367313564083578
Cr3: 0x1ae000
KdCopyDataBlock: 0xfffff807afdb04e4
KdDebuggerData: 0xfffff807b0601060
KdpDataBlockEncoded: 0xfffff807b06663e8
Current date/time: [2026-10-17 (YYYY-MM-DD) 17:22:36 (UTC)]
+ Processing...


Run bulk_extractor
Required Parameters
Scan: Image File Raw Device Directory of Files
Image file E:¥win7-32-nromanoff-c-drive.E01
Output Feature Directory D:¥output¥nromanoff2
General Options
Use Banner File
Use Alert List File
Use Stop List File
Use Find Regex Text File
Use Find Regex Text
Use Random Sampling
Tuning Parameters
Use Context Window Size 16
Use Page Size 16777216
Use Margin Size 4194304
Use Block Size 512
Use Number of Threads 4
Use Maximum Recursion Depth 7
Use Wait Time 60
Parallelizing
Use start processing at offset
Use process range offset o1-o2
Use add offset to reported feature offsets
Debugging Options
Scanners
base16
facebook
outlook
sceadan
wordlist
xor
accts
aes
base64
elf
email
exif
find
gps
gzip
hiberfile
httplogs
json
kml
msxml
net
ntfsindx
ntfslogfile
ntfsmft
ntfsusn



RESULT

Thus, the procedure for acquiring RAM using DumpIt and examining the captured memory for email-related digital evidence was studied.

EXPERIMENT 9

HIDING AND EXTRACTING A TEXT FILE BEHIND AN IMAGE/AUDIO FILE USING COMMAND PROMPT

AIM

To demonstrate the process of appending a text file to an image file and an audio file using Windows Command Prompt, with WinRAR used to prepare the archive for the audio-file method.

SOFTWARE REQUIREMENTS

- **Operating System:** Windows
- **Utility:** Command Prompt (CMD)
- **Software:** WinRAR for the audio-file procedure
- **Image File:** B.jpeg
- **Audio File:** sound.mp3
- **Text Files:** A.txt and secret.txt

THEORY

File concatenation is a method of joining the contents of two files to produce a combined output. In this experiment, Command Prompt is used to append a text file to an image file. In the audio-based method, the text file is first compressed into a ZIP archive using WinRAR, after which the archive is appended to an audio file through Command Prompt.

The resulting output contains the original image or audio data followed by the additional appended content. This technique demonstrates a basic method of concealing information within a combined file, although it does not provide encryption or guarantee that the added data will remain undetected.

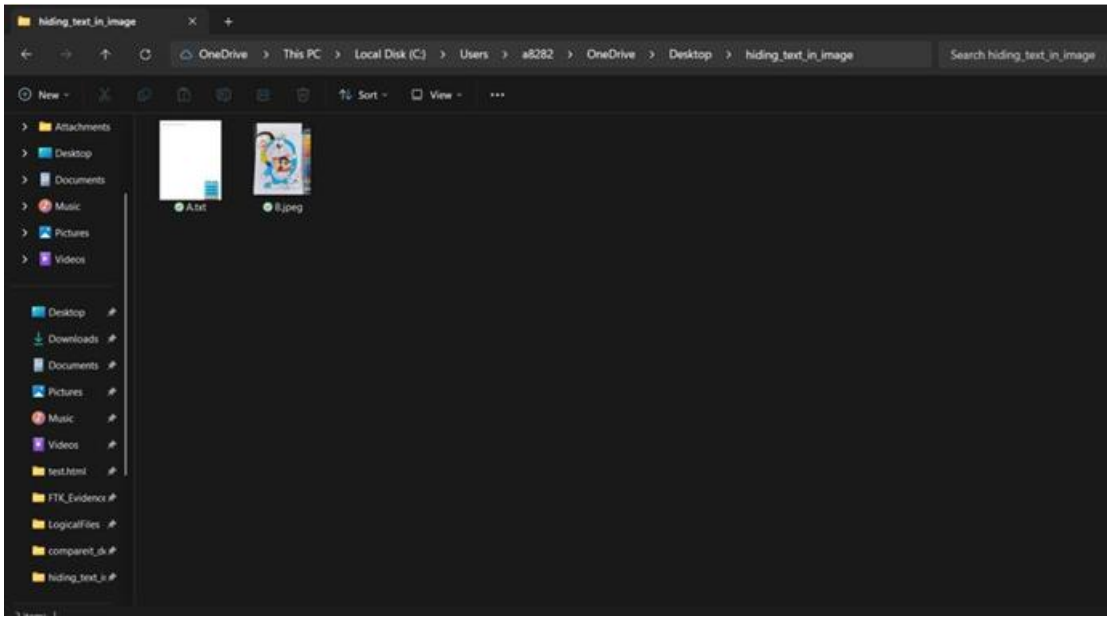
During digital forensic investigations, examiners can inspect file structures and identify data present beyond the expected file content. Such examination may help determine whether additional information has been appended and whether the concealed data can be recovered.

PROCEDURE

A. APPENDING A TEXT FILE TO AN IMAGE USING CMD

1. Create a folder named hiding_text_in_image on the Desktop.
2. Copy the image file B.jpeg and the text file A.txt into the newly created folder.
3. Open Command Prompt on the Windows computer.
4. Navigate to the working directory containing both files.
5. Execute the following command to combine the image and text file:

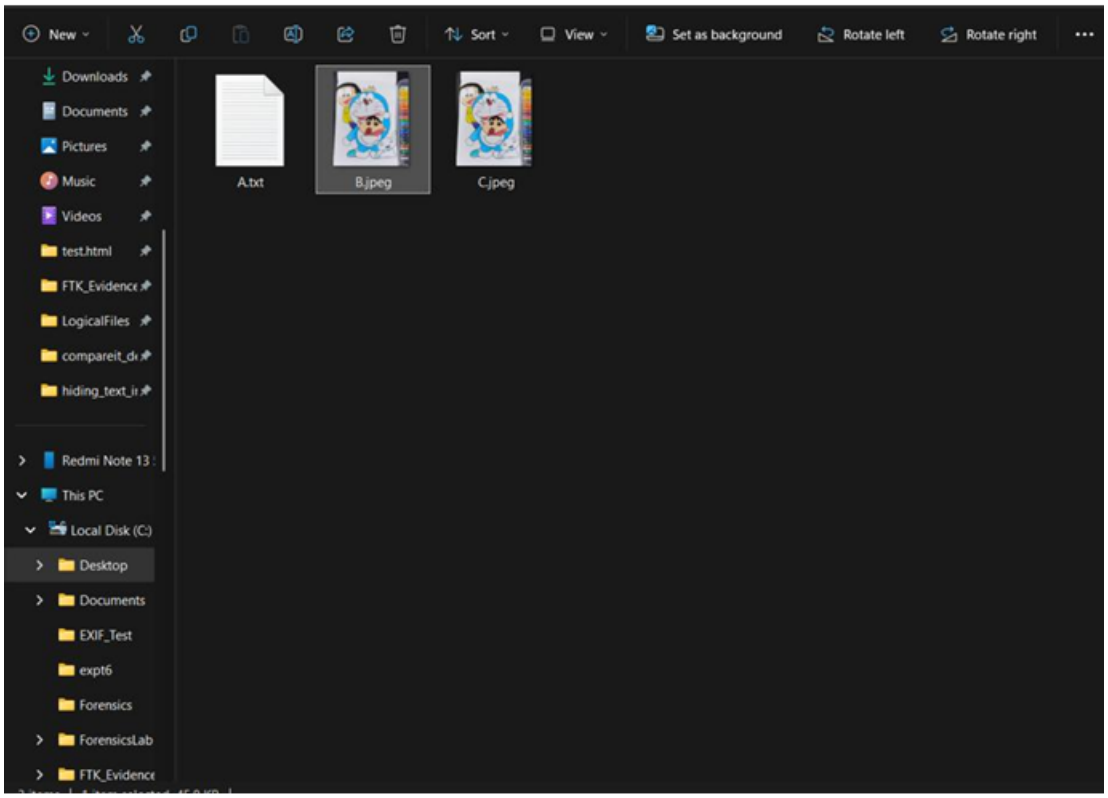
copy /b B.jpeg + A.txt C.jpeg



Description of the command:

- B.jpeg — The original image file.
- A.txt — The text file whose contents are appended.
- C.jpeg — The resulting combined output file.
- /b — Specifies binary mode for the file-copy operation.

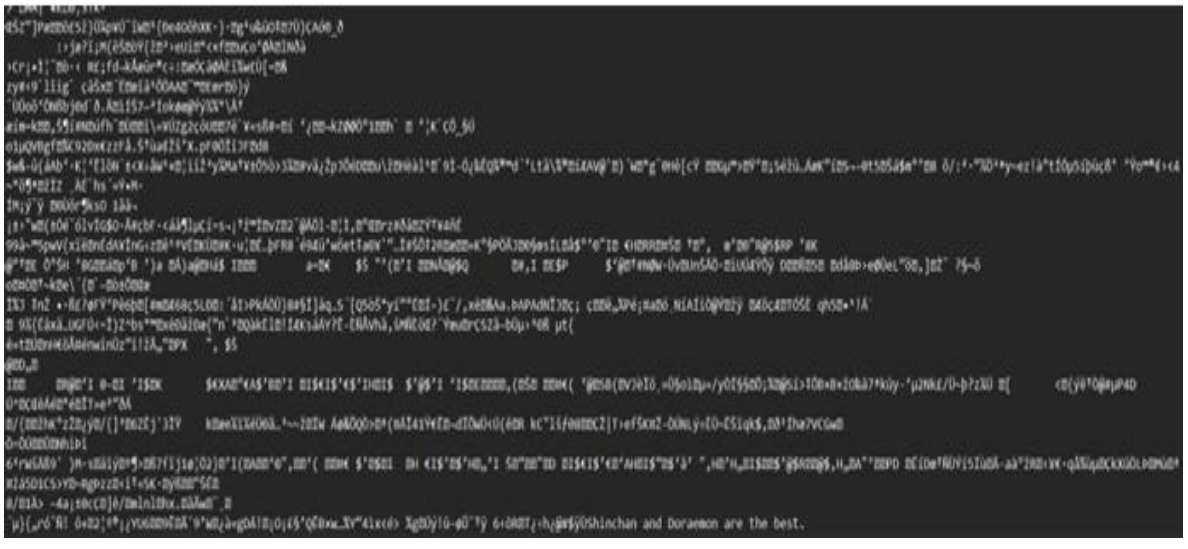
6. Wait until the command finishes executing.
7. Check the working directory to confirm whether C.jpeg has been created.
8. Record the command output and capture a screenshot for documentation.




```
C:\Desktop>cd hiding_text_in_image

C:\Desktop\hiding_text_in_image>copy /b B.jpeg + A.txt C.jpeg
B.jpeg
A.txt
        1 file(s) copied.

C:\Desktop\hiding_text_in_image>
```



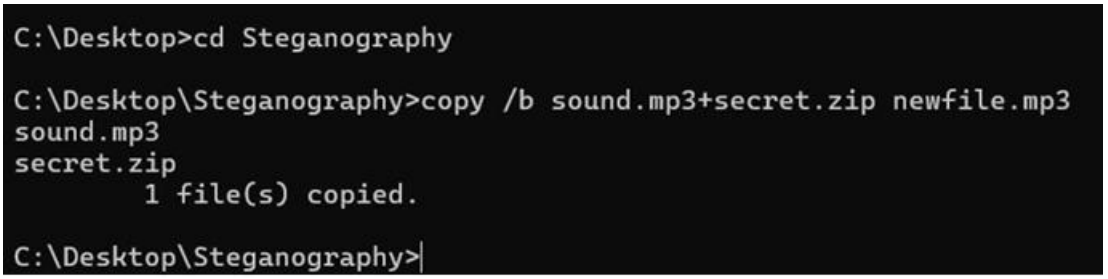
B. APPENDING A TEXT FILE TO AN AUDIO FILE USING WINRAR AND CMD

1. Open Notepad and create a text document named secret.txt.
2. Enter the required sample text and save the document.
3. Select secret.txt and use WinRAR to create a ZIP archive named secret.zip.
4. Place sound.mp3 and secret.zip in the same working directory.
5. Open Command Prompt and navigate to the folder containing the required files.
6. Run the following command to append the ZIP archive to the audio file:

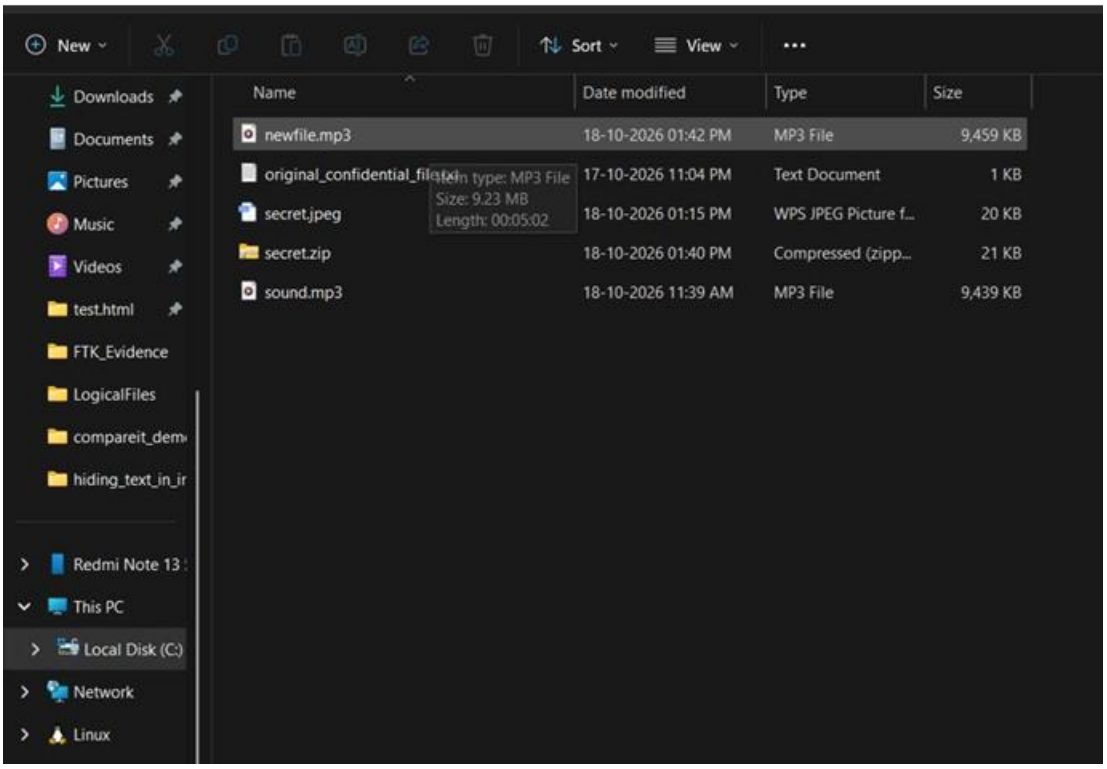
copy /b sound.mp3+secret.zip newfile.mp3

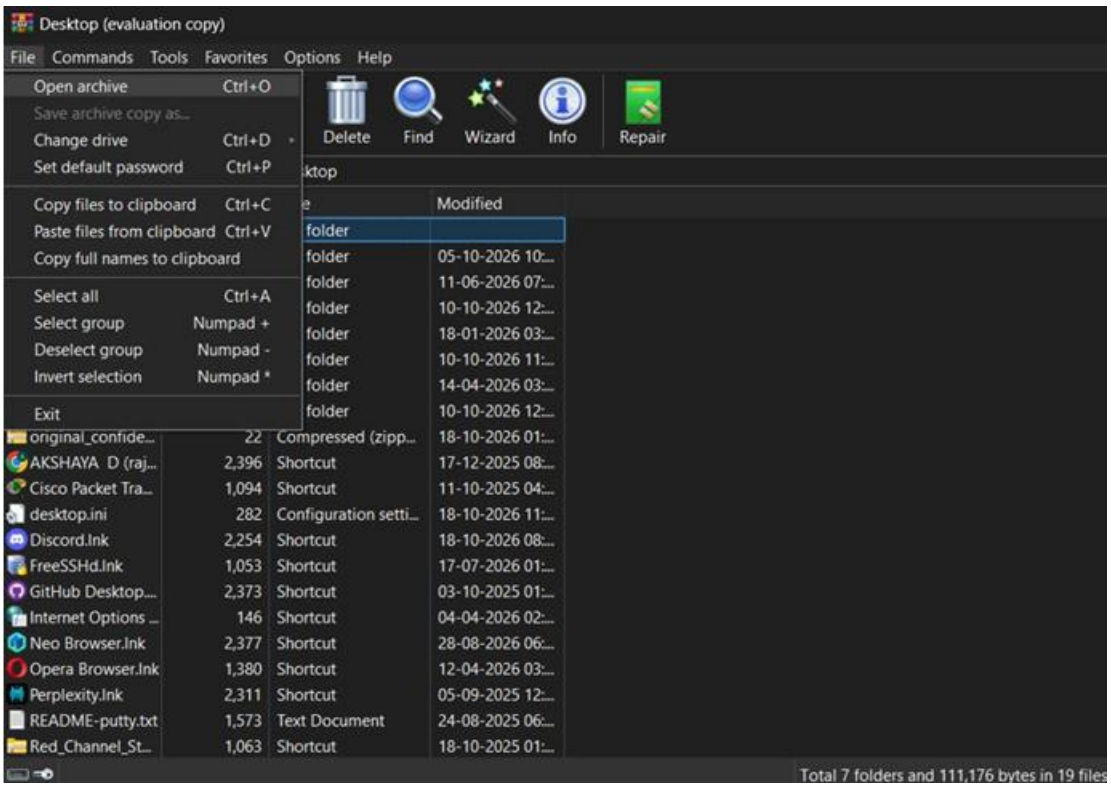
Description of the command:

- sound.mp3 — The original audio file.
- secret.zip — The compressed archive containing the text document.
- newfile.mp3 — The combined output file.
- /b — Enables binary-mode copying.

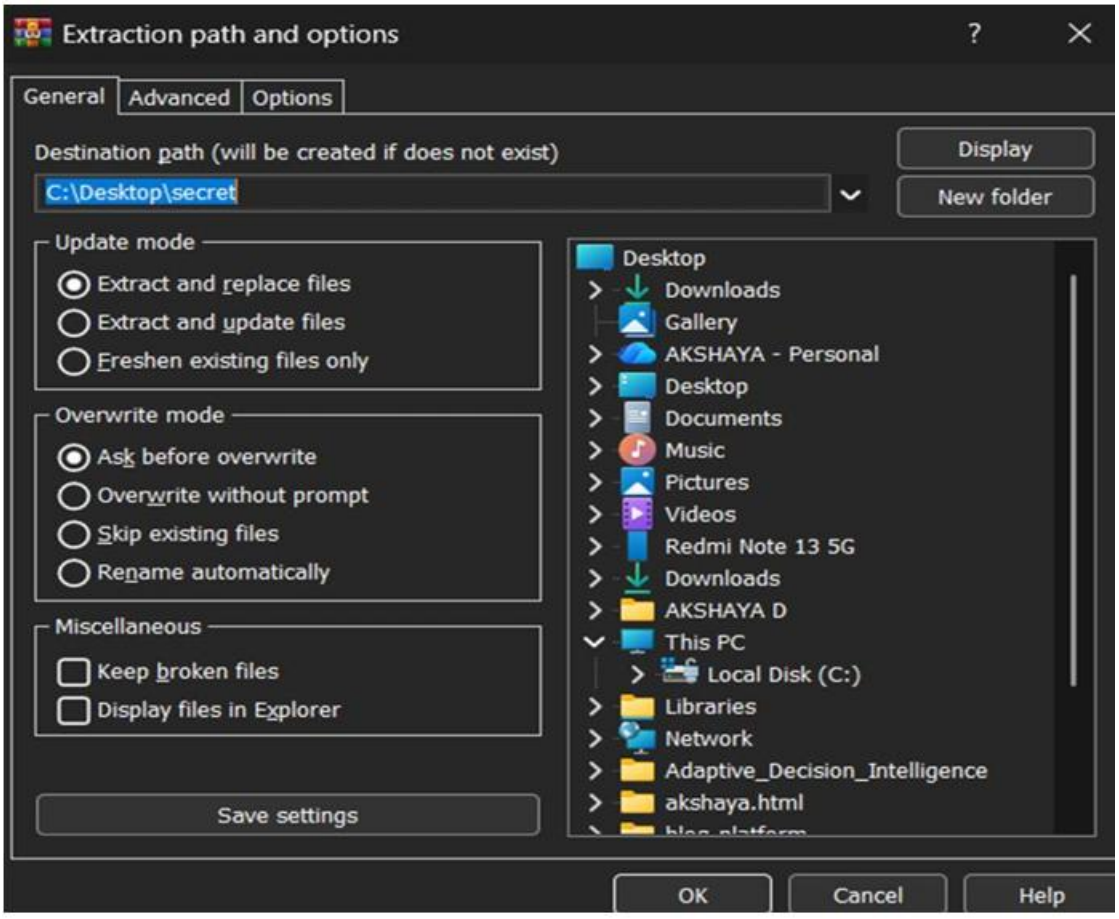


7. Allow the command to complete its execution.
8. Verify whether newfile.mp3 has been generated in the working directory.
9. If required by the lab procedure, open the output file in Notepad to observe its raw or binary-oriented content.
10. To attempt extraction, create a separate copy of newfile.mp3 and try opening that copy with WinRAR.
11. If WinRAR recognizes the appended ZIP archive, extract its contents and check whether secret.txt can be recovered.


New
Sort
View
Downloads
Documents
Pictures
Music
Videos
test.html
FTK_Evidence
LogicalFiles
compareit_dem
hiding_text_in_ir
Redmi Note 13
This PC
Local Disk (C:)
Network
Linux
Name
Date modified
Type
Size
newfile.mp3
18-10-2026 01:42 PM
MP3 File
9,459 KB
original_confidential_file.txt
17-10-2026 11:04 PM
Text Document
1 KB
Item type: MP3 File
Size: 9.23 MB
Length: 00:05:02
secret.jpeg
18-10-2026 01:15 PM
WPS JPEG Picture f...
20 KB
secret.zip
18-10-2026 01:40 PM
Compressed (zipp...
21 KB
sound.mp3
18-10-2026 11:39 AM
MP3 File
9,439 KB


Desktop (evaluation copy)
File
Commands
Tools
Favorites
Options
Help
Open archive Ctrl+O
Save archive copy as...
Change drive Ctrl+D
Set default password Ctrl+P
Copy files to clipboard Ctrl+C
Paste files from clipboard Ctrl+V
Copy full names to clipboard
Select all Ctrl+A
Select group Numpad +
Deselect group Numpad -
Invert selection Numpad *
Exit
Delete
Find
Wizard
Info
Repair
Modified
folder
05-10-2026 10:...
11-06-2026 07:...
10-10-2026 12:...
18-01-2026 03:...
10-10-2026 11:...
14-04-2026 03:...
10-10-2026 12:...
original_confide... 22 Compressed (zipp... 18-10-2026 01:...
AKSHAYA D (raj... 2,396 Shortcut 17-12-2025 08:...
Cisco Packet Tra... 1,094 Shortcut 11-10-2025 04:...
desktop.ini 282 Configuration setti... 18-10-2026 11:...
Discord.lnk 2,254 Shortcut 18-10-2026 08:...
FreeSSHd.lnk 1,053 Shortcut 17-07-2026 01:...
GitHub Desktop... 2,373 Shortcut 03-10-2025 01:...
Internet Options ... 146 Shortcut 04-04-2026 02:...
Neo Browser.lnk 2,377 Shortcut 28-08-2026 06:...
Opera Browser.lnk 1,380 Shortcut 12-04-2026 03:...
Perplexity.lnk 2,311 Shortcut 05-09-2025 12:...
README-putty.txt 1,573 Text Document 24-08-2025 06:...
Red_Channel_St... 1,063 Shortcut 18-10-2025 01:...
Total 7 folders and 111,176 bytes in 19 files



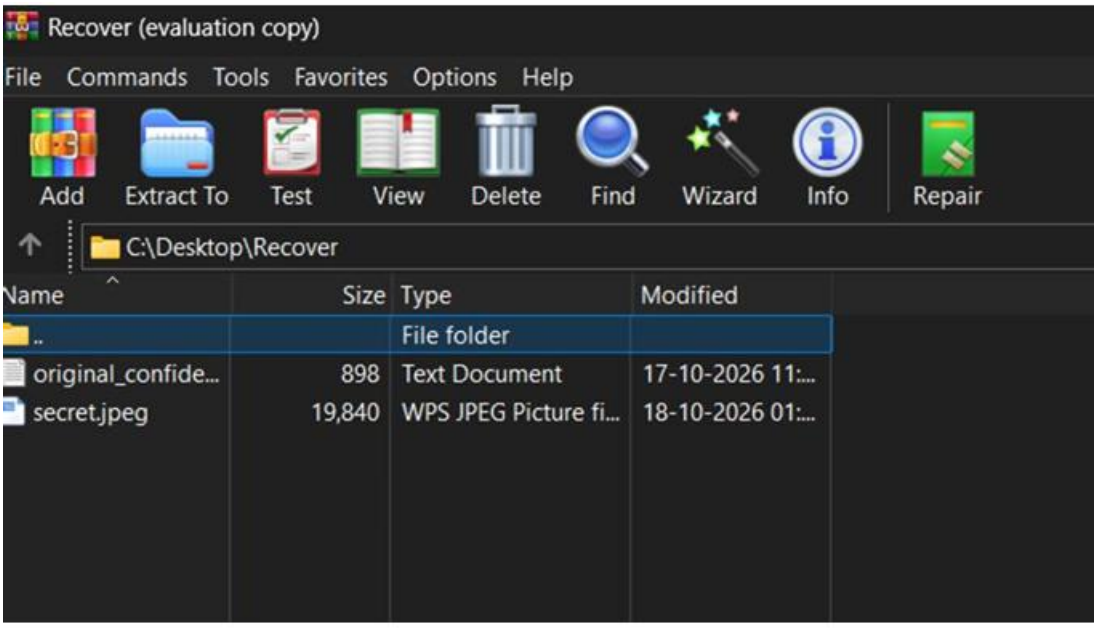


OBSERVATION

The binary copy commands were used to append text data to an image file and a ZIP archive to an audio file. The resulting files were named C.jpeg and newfile.mp3, respectively.

The experiment demonstrated how additional data can be placed after the original file content.

The possibility of extracting the appended text depended on whether the additional data remained intact and whether the archive could be recognized by the extraction utility.



RESULT

Thus, a text file was appended to an image file using Command Prompt, producing C.jpeg. A ZIP archive containing a text file was also appended to an audio file using WinRAR and Command Prompt, producing newfile.mp3. The procedure demonstrated file concatenation and the possibility of recovering appended data for forensic examination.

EXPERIMENT 10

EXTRACTION OF EXIF METADATA FROM IMAGE FILES USING EXIFREADER

AIM

To retrieve and analyze Exchangeable Image File Format (EXIF) metadata embedded in an image file with the help of ExifReader software.

SOFTWARE REQUIREMENTS

- Operating System: Windows
- Software Tool: ExifReader
- Input File: An image in JPEG or JPG format

THEORY

Exchangeable Image File Format (EXIF) is a metadata standard used to store additional information about digital images. Such metadata may be generated by digital cameras, smartphones, and other image-capturing devices when a photograph is taken.

EXIF metadata may include the following details:

- **Camera Make and Model:** Identifies the manufacturer and model of the device used to capture the image, when recorded.
- **Date and Time:** Provides the recorded date and time associated with the image.
- **Camera Settings:** Contains photographic parameters such as shutter speed, aperture, and exposure information.
- **Flash Status:** Indicates whether the flash was activated during image capture, if this information is available.
- **Image Dimensions:** Specifies the width and height of the image.
- **GPS Coordinates:** May provide the geographical location where the image was captured, provided that location data was recorded and retained.

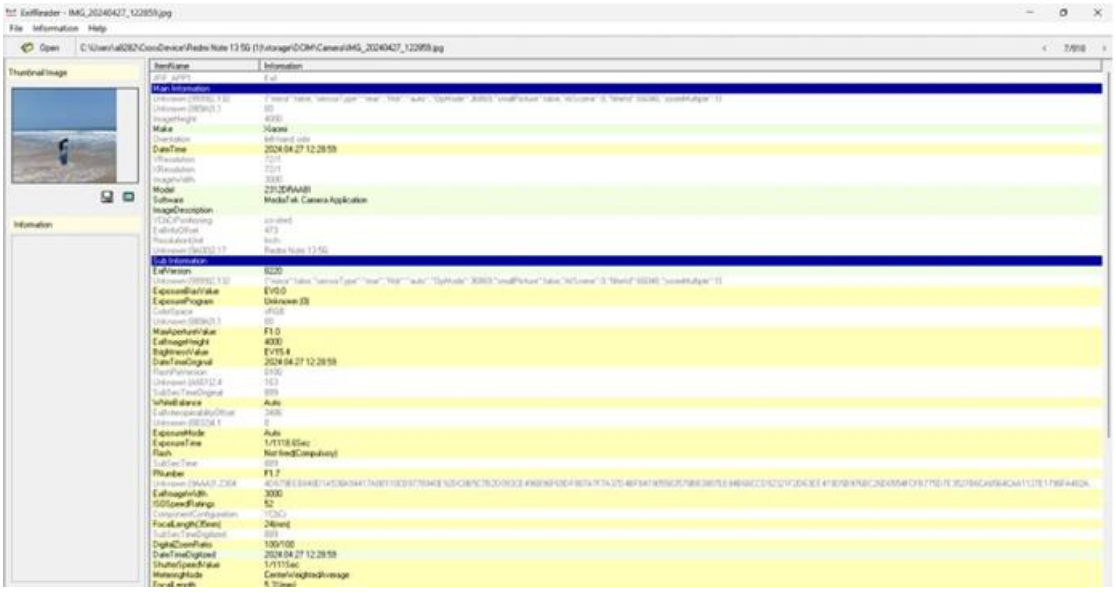
ExifReader is a Windows-based utility that enables users to view the EXIF metadata contained in supported image files. In digital forensics, this information can assist investigators in examining image properties and identifying potentially relevant details about the image's origin and capture conditions. However, the availability and accuracy of the metadata depend on the contents of the file.

PROCEDURE

1. Download and prepare the ExifReader utility on a Windows computer.
2. Run the ExifRead.exe application to open the software.
3. Click the **OPEN** option available in the application window.
4. Navigate to the folder containing the image file selected for examination.
5. Choose the required JPEG or JPG file and open it.
6. Verify that the selected image appears in the application.
7. Inspect the EXIF metadata presented by ExifReader.
8. Examine the available information, including camera make and model, image dimensions, date and time, and camera settings.
9. Note down the relevant metadata for further forensic analysis.
10. Take a screenshot of the application displaying the selected image and its available EXIF details for documentation.

OBSERVATION

ExifReader was used to display the selected image and examine the EXIF metadata present in the file. The information available for inspection depended on the metadata stored in the image.



RESULT

Thus, EXIF metadata was extracted and examined from the selected image file using ExifReader, enabling the identification of the image information available for digital forensic investigation.